

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination-2023

Course: B. Tech. Branch : AI&ML/AI&DS/AI

Semester :VII

Subject Code & Name: BTAIC702 Advanced Computer Vision

Max Marks: 60

Date:04-01-24

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Explain the concept of image representation and its importance in computer vision.	L2/CO1	6
B) Describe the process of contrast stretching in image enhancement.	L2/CO1	6
C) What is convolution in the context of image processing?	L2/CO1	6
Q.2 Solve Any Two of the following.		12
A) Explain the concept of noise models and their importance in image restoration.	L2/CO2	6
B) Differentiate between DFT (Discrete Fourier Transform) and DCT (Discrete Cosine Transform) in image processing.	L2/CO2	6
C) Demonstrate the process of scaling an image by a factor of 2 using a specific algorithm.	L3/CO2	6
Q. 3 Solve Any Two of the following.		12
A) Explain the concept of the Gray Level Co-occurrence Matrix (GLCM) in texture analysis.	L2/CO3	6
B) Identify the principles behind Prewitt, Sobel, and Canny edge detection methods.	L2/CO3	6
C) Describe the differences between region-based segmentation and edge-based segmentation.	L2/CO3	6

Q.4 Solve Any Two of the following.		12
A) Explain the concept of feature extraction and why it's essential in the process of building a dataset for visual recognition tasks.	L2/CO4	6
B) How does SIFT achieve scale invariance in feature detection and matching?	L2/CO4	6
C) Compare and contrast SVM, KNN, and Random Forest.	L4/CO4	6
Q. 5 Solve Any Two of the following.		12
A) Define Convolutional Neural Network (CNN) and its role in image classification.	L1/CO5	6
B) Describe the role of a generator and a discriminator in a GAN.	L2/CO5	6
C) Explain the multi-stage architecture of Fast R-CNN and how it improves upon R-CNN.	L2/CO5	6

***** End *****